CHEMICAL REACTIONS AND EQUATIONS

Class 10 Science — Chapter 1 Notes

1. Introduction to Chemical Reactions

A **chemical reaction** is a process in which one or more substances (reactants) react to transform into completely new substances (products) with entirely different physical and chemical properties. During a chemical reaction, rearrangement of atoms takes place, but atoms of one element do not change into those of another element.

- **Reactants:** The substances which take part in a chemical reaction. They are written on the left-hand side (LHS).
- **Products:** The new substances formed as a result of a chemical reaction. They are written on the right-hand side (RHS).

Characteristics of Chemical Reactions

A chemical reaction can be identified or confirmed by observing any of the following changes:

1. **Evolution of a Gas:** Reaction between zinc granules and dilute sulphuric acid.

$$Zn\ (s) + H_2SO_4\ (aq) \rightarrow ZnSO_4\ (aq) + H_2\ (g) \uparrow$$

2. **Formation of a Precipitate:** Mixing potassium iodide and lead nitrate solution forms a yellow precipitate of lead iodide.

$$Pb(NO_3)_2\ (aq) + 2KI\ (aq) \rightarrow PbI_2\ (s) \downarrow + 2KNO_3\ (aq)$$

3. **Change in Color:** Citric acid reacts with potassium permanganate solution (purple), causing its color to fade and become colorless.
4. **Change in Temperature:** Quicklime reacts vigorously with water to form slaked lime, releasing huge amounts of heat energy (Exothermic).

$$CaO\ (s) + H_2O\ (l) \rightarrow Ca(OH)_2\ (aq) + Heat$$

5. **Change in State:** Burning of a wax candle (Solid to liquid and gas).

2. Chemical Equations and Balancing

A **chemical equation** is a symbolic shorthand representation of a chemical reaction using symbols and chemical formulae of the substances involved.

Law of Conservation of Mass

According to this law, *mass can neither be created nor destroyed in a chemical reaction*. Consequently, the total mass of elements present in the products must be equal to the total mass of elements present in the reactants. Thus, the number of atoms of each element must remain the same before and after a reaction.

Types of Equations

- **Skeletal (Unbalanced) Equation:** The number of atoms of various elements is not equal on both sides. Example: $Mg + O_2 \rightarrow MgO$
- **Balanced Equation:** The number of atoms of each element is equal on both sides. Example: $2Mg + O_2 \rightarrow 2MgO$

> **Note on Balancing:** We balance a chemical equation by multiplying chemical formulas by coefficients (coefficients go in front, e.g., $2H_2$). We never change the subscripts within a formula (e.g., H_2O cannot be changed to H_2O_2).

3. Types of Chemical Reactions

I. Combination Reaction

A reaction in which two or more reactants combine together to form a **single product**.

- **Burning of coal:** $C (s) + O_2 (g) \rightarrow CO_2 (g)$
- **Formation of water:** $2H_2 (g) + O_2 (g) \rightarrow 2H_2O (l)$

II. Decomposition Reaction

A reaction in which a single reactant breaks down into two or more simpler substances. These reactions require energy in the form of heat, light, or electricity, and are opposite to combination reactions.

1. **Thermal Decomposition** Heat**:** Carried out by heating.

$$2FeSO_4 (s)\ [Green]\ \&xrightarrow;\Delta\ Fe_2O_3 (s)\ [Brown] + SO_2 (g) + SO_3 (g)$$

$$CaCO_3 (s)\ [Limestone]\ \&xrightarrow;\Delta\ CaO (s)\ [Quicklime] + CO_2 (g)$$

2. **Electrolytic Decomposition** Electricity **:** Carried out by passing electricity (Electrolysis).

$2H_2O$ *(l)* &xrightarrow;*Electricity* $2H_2$ *(g)* + O_2 *(g)*

3. **Photolytic Decomposition** Light **:** Carried out in the presence of sunlight.

2AgCl (s) [White] &xrightarrow;*Sunlight 2Ag (s) [Grey]* + Cl_2 *(g)*

**Application: This reaction is utilized in black and white photography.*

III. Displacement Reaction

A chemical reaction in which a more reactive element displaces a less reactive element from its salt solution.

- **Iron nail in Copper Sulphate solution:** The blue color of copper sulphate solution fades to light green due to the formation of iron sulphate, and a brown coating of copper deposits on the iron nail.

$$Fe\ (s) + CuSO_4\ (aq)\ [Blue] \rightarrow FeSO_4\ (aq)\ [Green] + Cu\ (s)$$

- **Zinc with copper sulphate:** $Zn\ (s) + CuSO_4\ (aq) \rightarrow ZnSO_4\ (aq) + Cu\ (s)$

IV. Double Displacement Reaction

A reaction in which two compounds react by an exchange of ions to form two new compounds. These often result in the formation of an insoluble precipitate.

- **Reaction between Sodium Sulphate and Barium Chloride:** A white precipitate of Barium Sulphate forms immediately.

$$Na_2SO_4\ (aq) + BaCl_2\ (aq) \rightarrow BaSO_4\ (s) \downarrow [White\ ppt] + 2NaCl\ (aq)$$

V. Oxidation and Reduction (Redox Reactions)

- **Oxidation:** The gain of oxygen or loss of hydrogen during a chemical reaction.
- **Reduction:** The loss of oxygen or gain of hydrogen during a chemical reaction.
- **Redox Reaction:** A reaction in which both oxidation and reduction occur simultaneously.

CuO + H_2 &xrightarrow;*Δ Cu* + H_2O

Here, **CuO** loses oxygen (Reduced to Cu) | **H_2** gains oxygen (Oxidized to H_2O)

4. Key Differentiations

Difference 1: Exothermic vs. Endothermic Reactions

Feature	Exothermic Reactions	Endothermic Reactions
Definition	Reactions in which heat energy is released into the surroundings along with products.	Reactions in which energy is absorbed from the surroundings to proceed.
Temperature Change	Causes a rise in the temperature of the container/surroundings.	Causes a decrease in the temperature of the surroundings.
Energy Sign	Represented with + ***Heat*** on the product side.	Represented with + ***Heat*** on the reactant side or over the arrow.
Key Examples	1. Respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$ 2. Burning of Natural Gas (CH_4).	1. Photosynthesis: $6CO_2 + 6H_2O + Sunlight \rightarrow C_6H_{12}O_6 + 6O_2$ 2. All decomposition reactions.

Difference 2: Displacement vs. Double Displacement Reactions

Feature	Displacement Reaction	Double Displacement Reaction
Core Mechanism	A more reactive element replaces a less reactive element from its compound.	Two atomic groups/ions mutually exchange places to form two new compounds.
Reactant Nature	Involves one highly reactive element and one stable salt solution compound.	Involves two ionic compounds interacting in an aqueous solution state.
Precipitation	No specific insoluble precipitates are characteristically formed.	Characteristically forms an insoluble precipitate in most instances.
Example	$Fe\ (s) + CuSO_4\ (aq) \rightarrow FeSO_4\ (aq) + Cu\ (s)$	$AgNO_3\ (aq) + NaCl\ (aq) \rightarrow AgCl\ (s)\downarrow + NaNO_3\ (aq)$

5. Effects of Oxidation in Daily Life

1. Corrosion

The gradual destruction or eating away of metals by the action of air, moisture, water, or chemical acids on their surface.

- **Rusting of Iron:** Iron reacts with oxygen and moisture to form a flaky brown substance called hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$).
- **Corrosion of Silver:** Silver reacts with sulfur compounds in air to form a black layer of Silver Sulphide (Ag_2S).
- **Corrosion of Copper:** Copper reacts with moist carbon dioxide to form a green coating of basic Copper Carbonate ($CuCO_3 \cdot Cu(OH)_2$).
- **Prevention Methods:** Painting, oiling, greasing, chrome plating, or **Galvanization** (coating a layer of zinc over iron surfaces).

2. Rancidity

The oxidation of fats and oils contained in food items when exposed to open air for a prolonged time, causing a change in their odor, appearance, and taste.

- **Prevention Methods:**
  - Adding food-safe **Antioxidants** (e.g., BHA - Butylated Hydroxyanisole, BHT).
  - Packaging snacks/chips in airtight containers.
  - Flushing product packaging bags with unreactive **Nitrogen Gas** to displace oxygen.
  - Storing food materials at low temperatures inside a refrigerator.